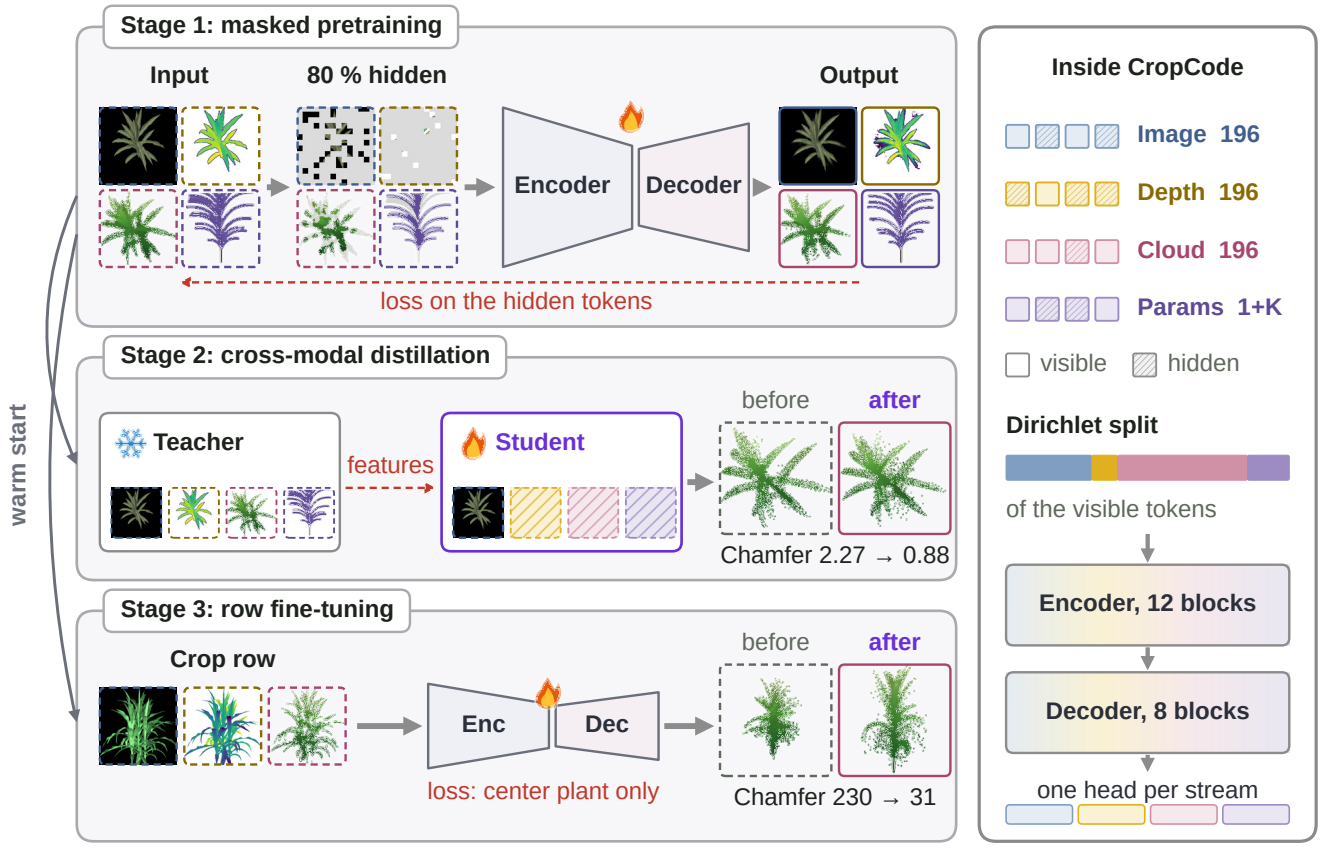

Stage 1: masked pretraining
Input
80 % hidden
Output
Encoder
Decoder
loss on the hidden tokens
Stage 2: cross-modal distillation
warm start
Teacher
features
Student
before
after
Chamfer 2.27 → 0.88
Stage 3: row fine-tuning
Crop row
Enc
Dec
loss: center plant only
before
after
Chamfer 230 → 31
Inside CropCode
Image 196
Depth 196
Cloud 196
Params 1+K
visible
hidden
Dirichlet split
of the visible tokens
Encoder, 12 blocks
Decoder, 8 blocks
one head per stream